

3(a)(i)	any three from: 1 no, nucleus / organelles ; ora e.g. 'if they were white blood cells ...' 2 cytoplasm is, homogeneous / AW ; 3 about same, size / width / diameter, as lumen of capillary ; 4 cells about 7 μm in diameter / AW ; 5 flexible / variety of shapes / irregular shapes ; 6 AVP ; e.g. rouleau	3
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Question	Answer	Marks
3(a)(ii)	<i>assume answers are about tissue fluid unless told otherwise</i> any two from: 1 no red blood cells ; 2 no platelets ; 3 fewer (named) protein(s) / no large proteins / no plasma proteins ; A no named plasma proteins (albumen / fibrinogen) A fewer plasma proteins only if stated that they are, small / leave blood 4 less, glucose / amino acids / fatty acids ; 5 fewer, white blood cells / leucocytes / neutrophils / monocytes ; A more macrophages 6 less / lower concentration of, oxygen or more / higher concentration of, carbon dioxide ; I urea except in liver and muscle tissue R 'waste'	2
3(b)(i)	<i>ignore any explanations</i> (mean) percentage saturation of haemoglobin with oxygen (in blood leaving the lungs) decreases ; (mean) haemoglobin concentration in blood increases ; any suitable comparative data quote with altitude in m and mean Hb concentration in g 100 cm⁻³ or percentage saturation ;	3

Question	Answer	Marks
3(b)(ii)	any three from: 1 more red blood cells ; 2 larger red blood cells ; 3 more haemoglobin per red blood cell ; 4 more alveoli / larger surface area for gas exchange ; A bigger lungs / larger chest volume / broader chest / larger capacity of lungs (total or vital) 5 higher, cardiac output / stroke volume / AW ; A higher blood pressure in pulmonary artery 6 AVP ; 7 AVP ; e.g. higher tidal volume / deeper breaths higher ventilation rate (minute volume) I faster breathing <i>ref. to</i> erythropoietin / EPO, stimulating production of (more) red blood cells more red blood cells through alveolar capillaries per unit time	3